DSM A2 Part 2 Final Report: MongoDB + Neo4j Querying

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April 6, 2026

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1 MongoDB Querying

Workflow and Execution

The full MongoDB Part 2 pipeline is implemented in `queries/p2_mongo.py`. It writes complete aggregation pipelines, executes all required queries, saves tabular outputs, and generates visualizations.

Run sequence:

```
source .venv/bin/activate
pip install -r requirements.txt
python queries/p2_mongo.py
```

Generated artifacts:

- Pipelines: `queries/mongo_part2_queries.txt`
- Data outputs: `report/p2/data/mongo/q*.csv`, `report/p2/data/mongo/q*.json`
- Figures: `report/p2/figures/mongo/q*.png`

Q1: Cohort Analysis of User Reviewing Behaviour

Cohorts are defined by year of `users.yelping_since`. For each cohort, the query computes mean and standard deviation of stars, mean review character length, mean useful votes per review, and star-bucket proportions.

Key outputs from `q1_summary.json`:

- Highest mean star cohort: **2005** (**3.903**)
- Highest mean useful-votes cohort: **2004** (**2.276**)
- Total cohorts: **19**

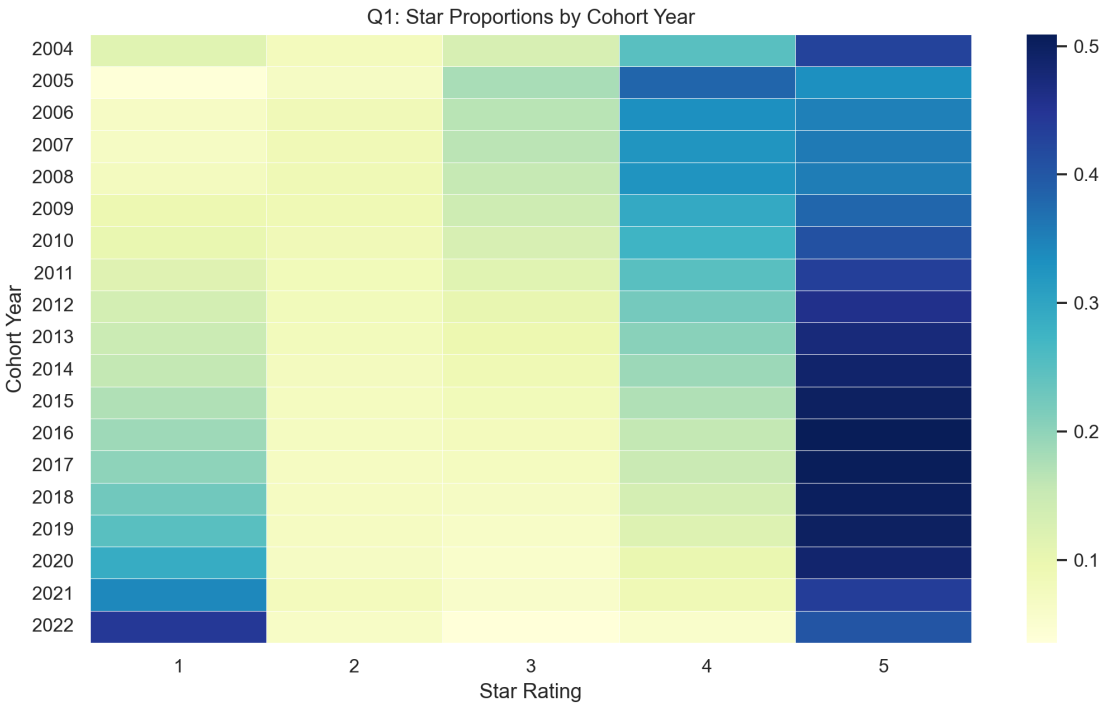

Figure 1: Q1 heatmap: star proportions by cohort year.

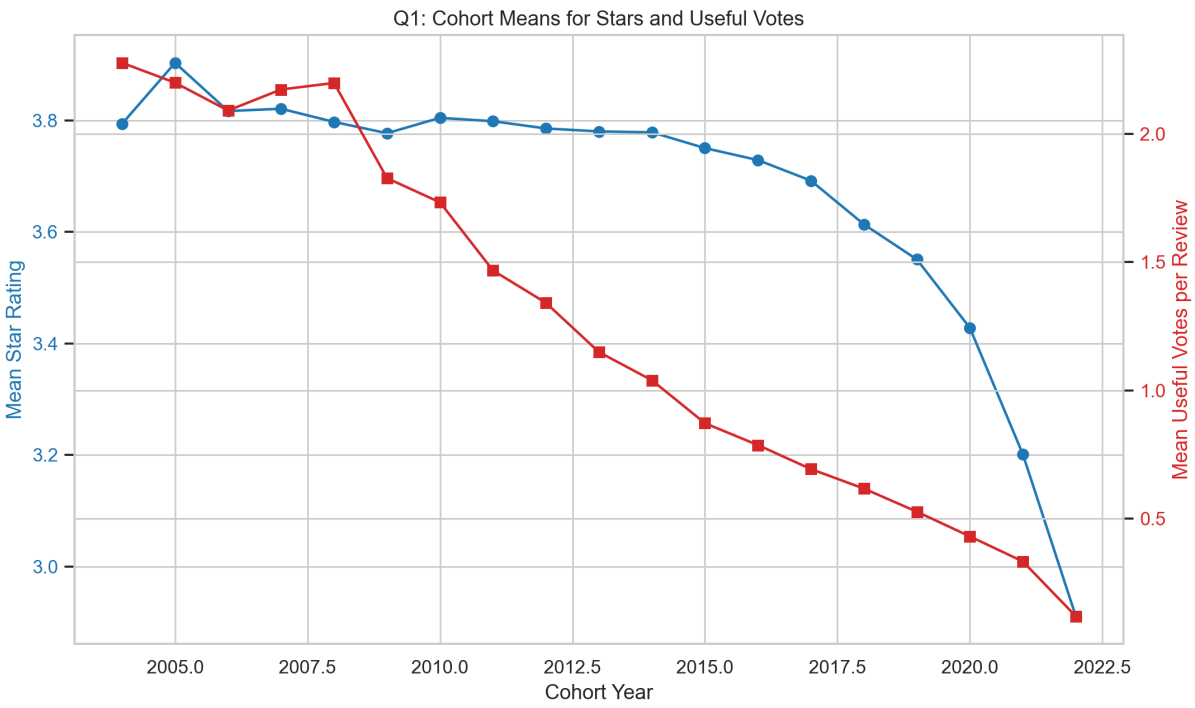

Figure 2: Q1 line plot: mean stars and mean useful votes by cohort year.

Interpretation:

- Older cohorts show stronger useful-vote and rating signals than recent cohorts.
- Later cohorts (notably post-2018) trend lower on both dimensions.

Q2: Month-over-Month Trend Consistency by Category

For each category with at least 500 reviews, monthly average stars are computed and consistency is measured as the proportion of consecutive month-pairs with increase/decrease.

Top 3 upward categories:

- Japanese Curry (0.544)
- Home Cleaning (0.543)
- Beer Gardens (0.541)

Top 3 downward categories:

- Medical Centers (0.566)
- Walk-in Clinics (0.554)
- Eyelash Service (0.550)

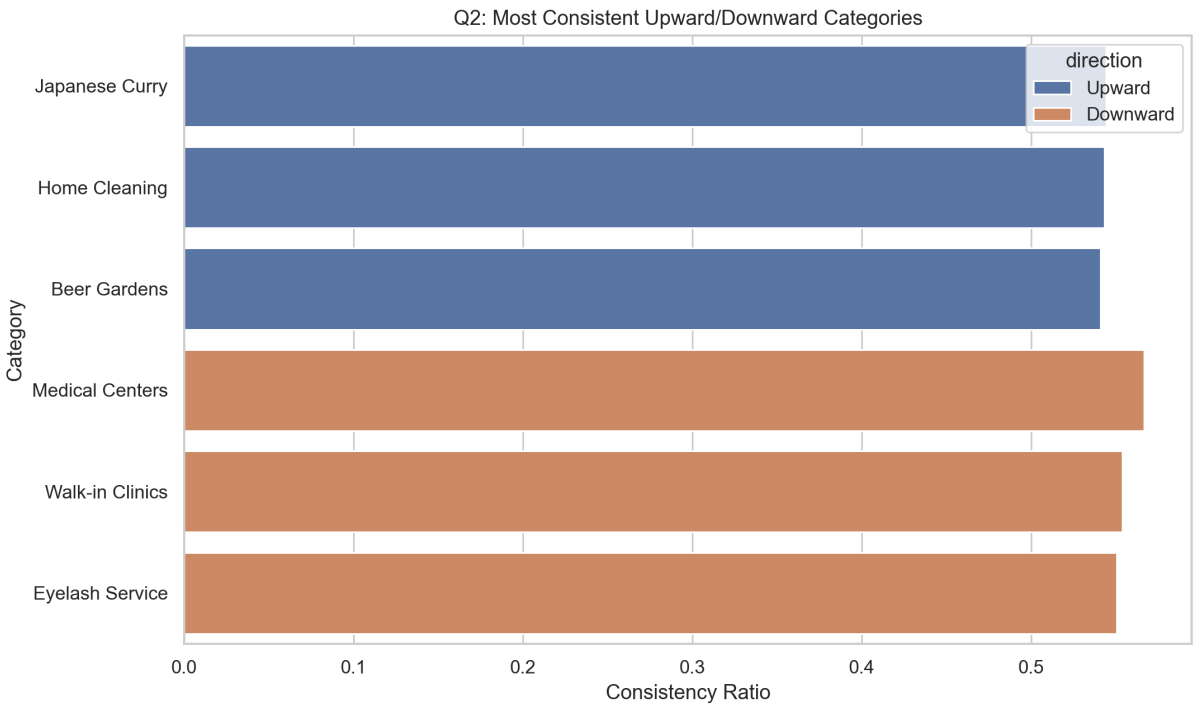

Figure 3: Q2 bar plot: consistency ratios for most upward/downward categories.

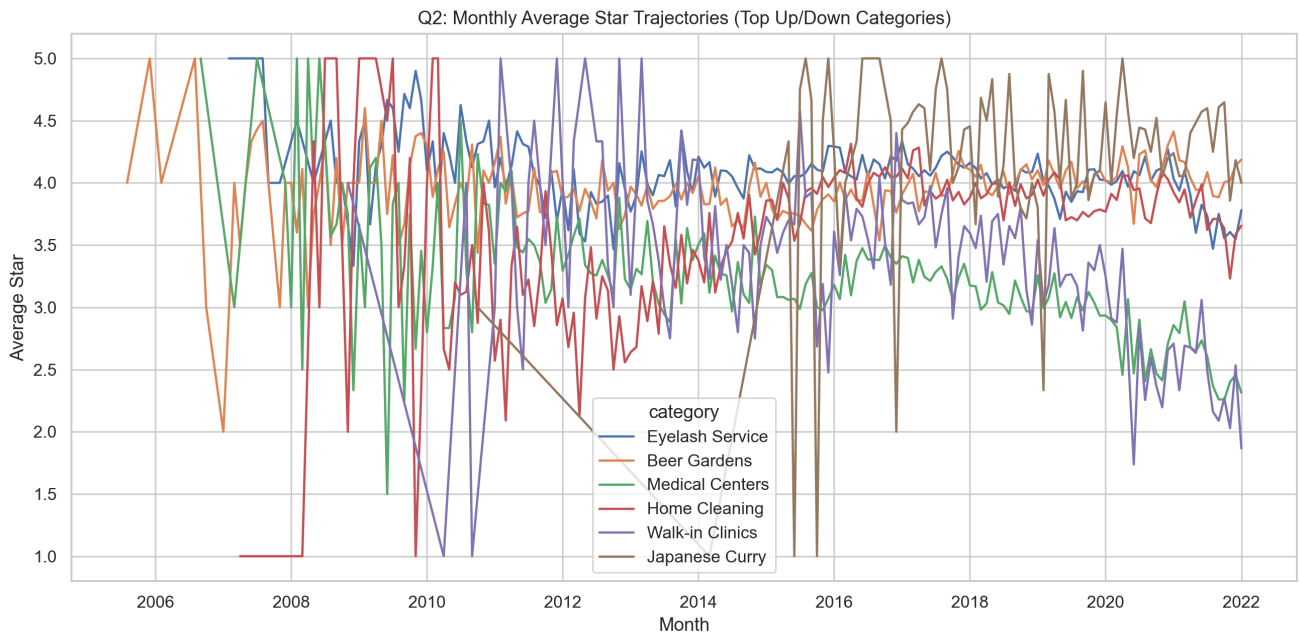

Figure 4: Q2 line plot: monthly trajectories for selected categories.

Interpretation:

- Directional consistency exists, but no category is fully monotonic.
- Both upward and downward groups show meaningful month-to-month volatility.

Q3: Check-in Class Cross-tabulation

Businesses are classified into low/medium/high check-in classes using quartiles from check-in counts (Q1=6, Q3=72). For each top-10 category by total review count, metrics are computed by class: mean stars, mean

review count, and tips-to-reviews ratio.

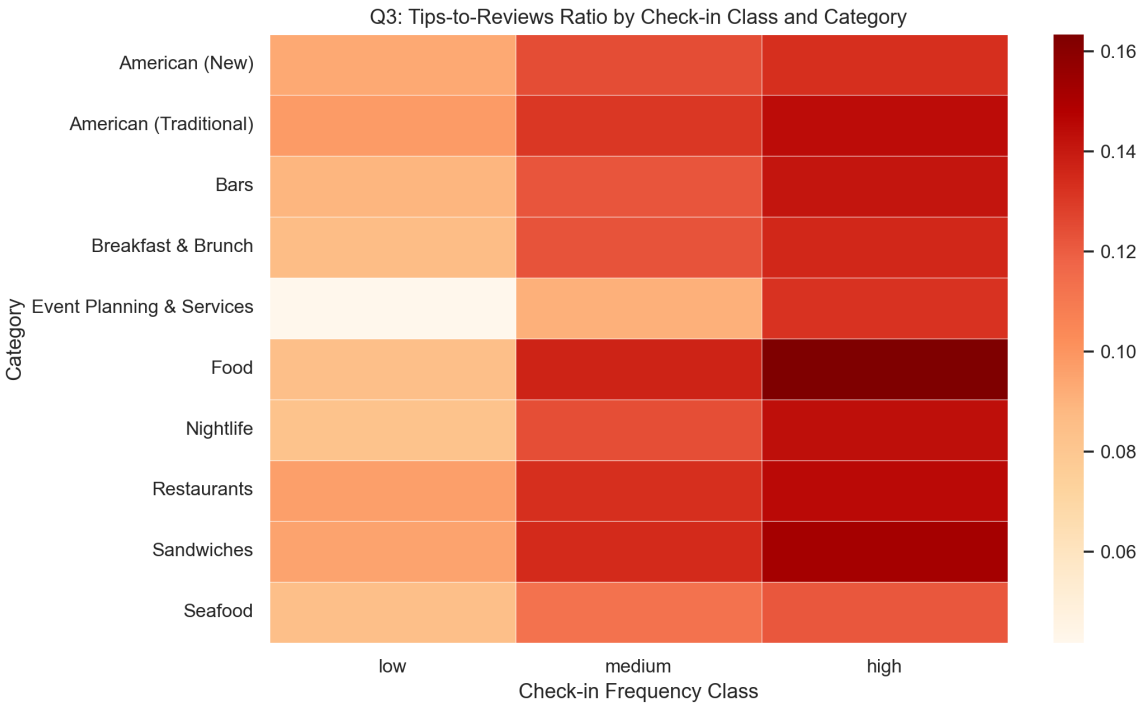

Figure 5: Q3 heatmap: tips-to-reviews ratio by category and check-in class.

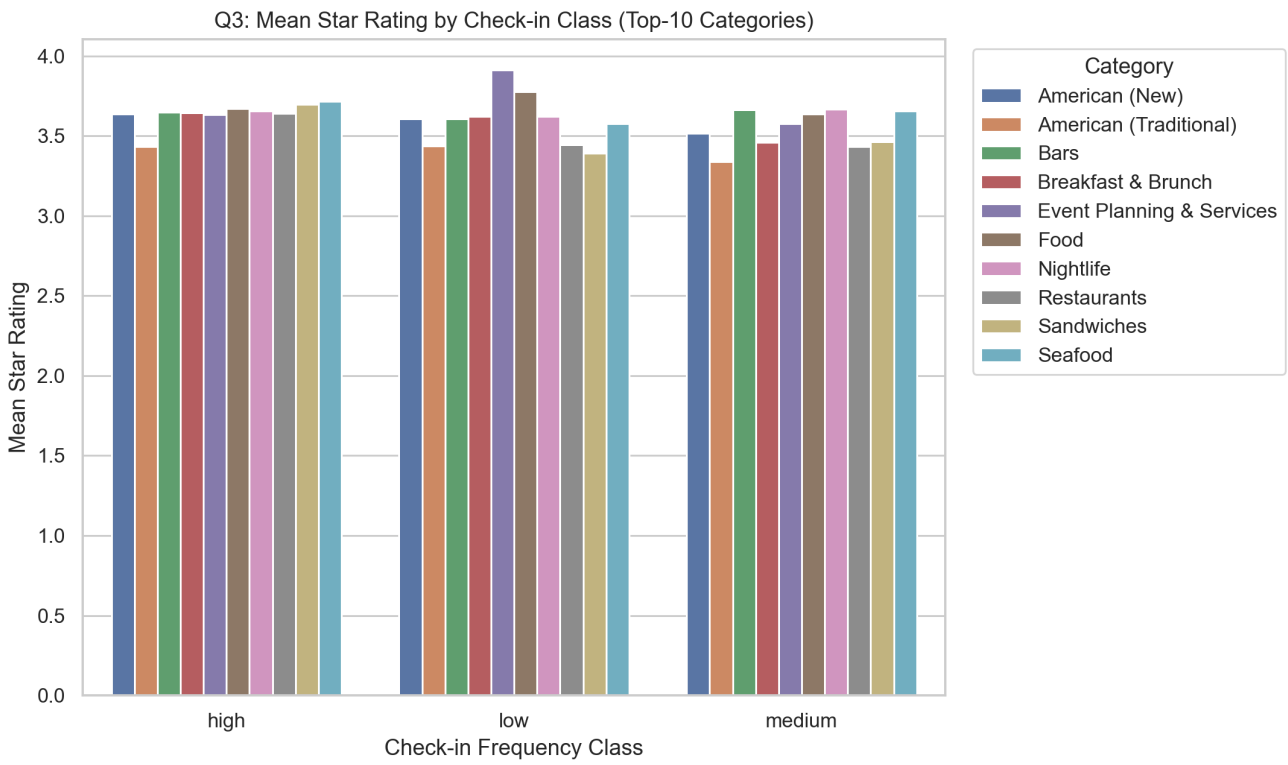

Figure 6: Q3 bar plot: mean stars by check-in class across top categories.

Interpretation:

- High check-in businesses generally have substantially higher mean review volumes.
- Tips-to-reviews intensity is commonly strongest in the high check-in group.

2 Shared Subset Construction (Used for Neo4j and Predictive Modelling)

Why a Shared Subset Was Built

Part 2 includes graph analytics and predictive modelling workloads that are expensive on the full Yelp dump. To keep execution feasible and reproducible on local hardware while preserving assignment complexity, a single materialized subset was built and then reused across both downstream tracks:

- Neo4j + GDS querying
- Predictive modelling dataset construction and evaluation

Using one canonical subset avoids dataset drift between tasks and ensures that model features, graph structures, and exploratory results are all computed from the same entity universe.

Subset Construction Pipeline (ETL)

The subset is created by `etl/build_shared_subset.py`. The logic is deterministic given parameters and follows these steps:

1. Scan businesses and keep only valid candidates (non-empty state/city/categories, minimum review count, optional check-in constraint).
2. Allocate a target business budget across top states using proportional quotas with a minimum floor per state.
3. Within each state, select businesses in a round-robin manner across primary categories to improve category diversity.
4. Keep reviews only for selected businesses, with a per-business cap.
5. Keep users referenced by selected reviews.
6. Prune each kept user's friend list to subset users only (keeps social edges coherent inside subset).
7. Keep check-ins and tips for selected businesses.

This produces a smaller but structurally consistent induced multi-collection/multi-graph dataset.

Subset Configuration and Resulting Size

From `data/subset_shared/subset_manifest.json` (subset name: `shared_v1`):

- **Parameters:** target businesses=6000, max states=12, min businesses/state=250, min business review count=15, max reviews/business=120.
- **Counts:** businesses=6000, reviews=437582, users=289869, check-ins=5634, tips=147910.

State quota allocation preserved geographic spread while still emphasizing high-signal states. Largest state allocations were PA (1250) and FL (1054), with additional representation from TN, IN, MO, LA, AZ, NV, NJ, CA, ID, and AB.

Why This Matters for the Report

All Neo4j and predictive modelling results reported below are executed on this shared subset namespace (`subset_shared`), not on the full original graph, to keep conclusions internally consistent.

3 Neo4j / Cypher Querying

NQ1: PageRank on User-Business Projection

The query projects a directed weighted graph from users to businesses using review stars as weights and runs GDS PageRank with minimum 20 iterations.

Deliverables produced by script:

- Top-15 businesses by PageRank with review count and average stars
- Spearman correlations: PageRank vs review-count ranking; PageRank vs average-star ranking
- Rank-gap analysis for businesses over/under-ranked by PageRank vs simple metrics

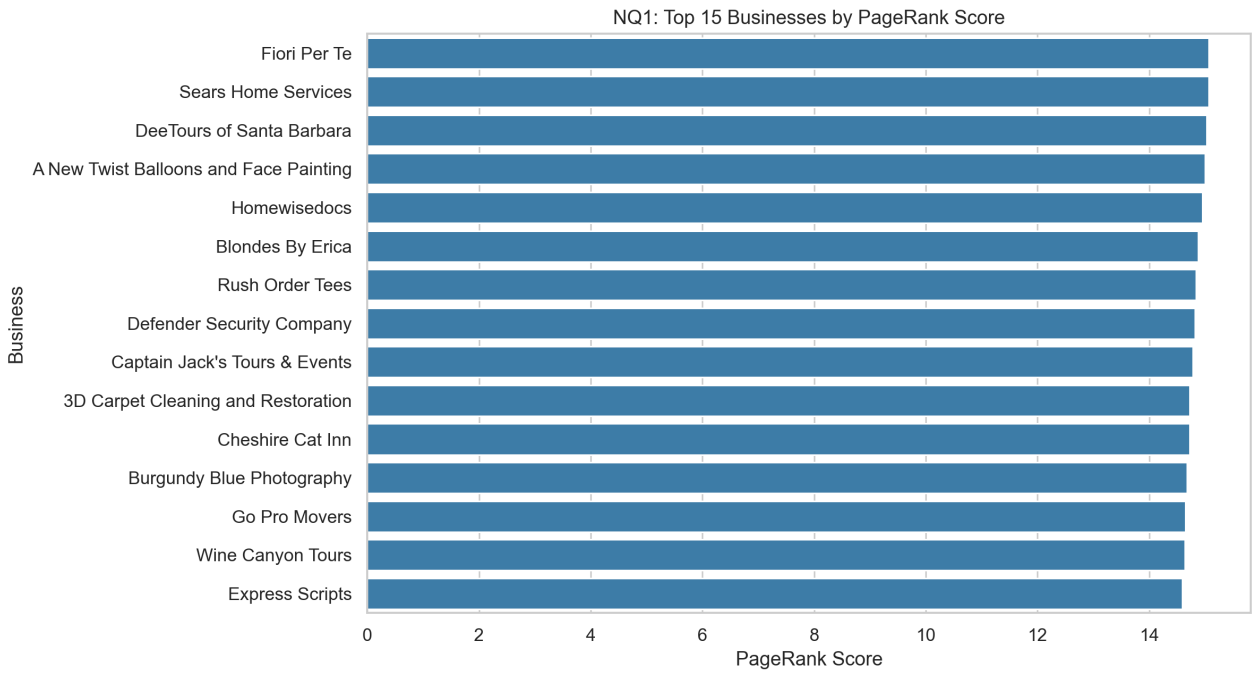

Figure 7: NQ1: Top 15 businesses by PageRank score.

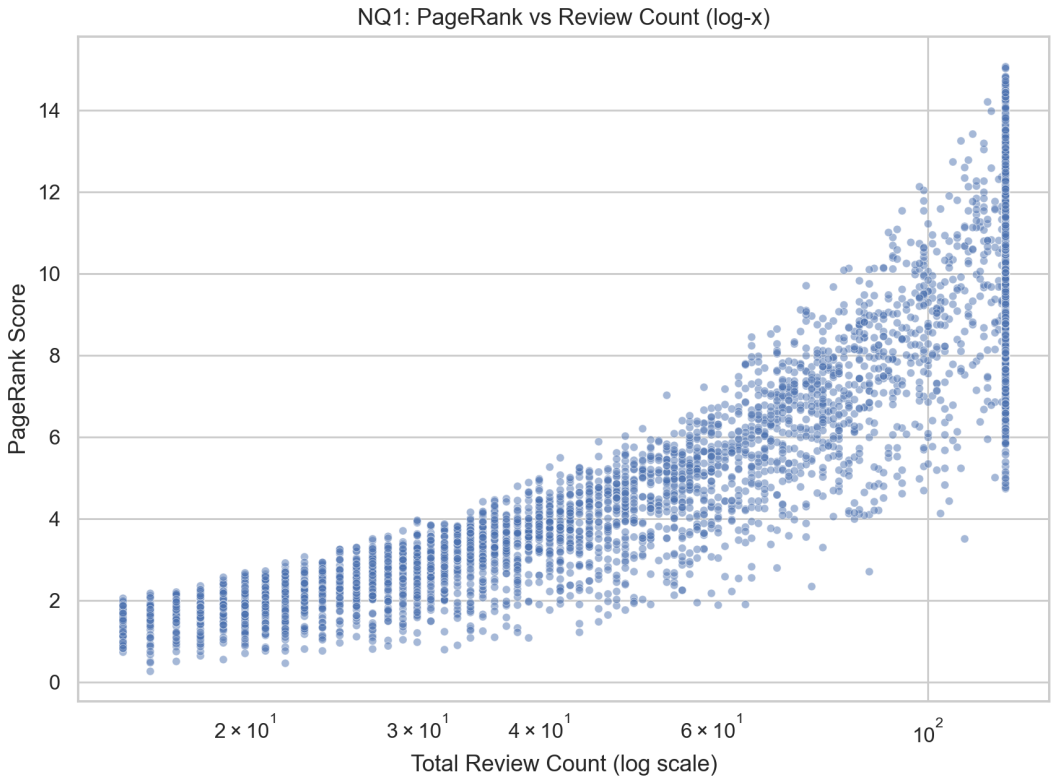

Figure 8: NQ1: PageRank score vs review count.

Key findings (`n1_summary.json`):

- Spearman correlation(PageRank, review count) = **0.929**, indicating strong agreement between authority and volume.
- Spearman correlation(PageRank, average stars) = **0.085**, suggesting weak monotonic dependence on mean rating.
- Top-ranked business by PageRank: **Fiori Per Te**.

Analytical interpretation:

- The very high PageRank vs review-count correlation indicates that in this subset, **exposure and interaction volume dominate** influence. Businesses with many inbound user interactions absorb centrality mass regardless of sentiment.
- The weak PageRank vs mean-stars correlation shows that **network prominence is not equivalent to customer satisfaction**. A business can be structurally central while being poorly rated, and vice versa.
- Rank-gap diagnostics reveal two distinct regimes:
  - **Over-ranked by PageRank** (large positive gap): entities such as Dent Artist Paintless Dent Removal and Better Business Bureau Serving West Florida have moderate review counts but high graph centrality, suggesting they sit in reviewer pathways that connect active user neighborhoods.
  - **Under-ranked by PageRank** (large negative gap): high-volume brands such as Whole Foods Market and Walmart Supercenter receive many reviews but are less central in reviewer flow, consistent with broad but less tightly connected audience overlap.

Business insights:

- For recommendation/ranking strategy, a two-signal policy is preferable: centrality for discoverability and rating for quality control.
- For business growth, partnerships with highly connected reviewer segments may improve visibility faster than marginal rating improvements alone.

NQ2: Louvain Community Detection

The script runs Louvain on the `FRIENDS_WITH` social graph, filters communities with at least 25 members, and computes top states, top categories, and geographic concentration index (GCI).

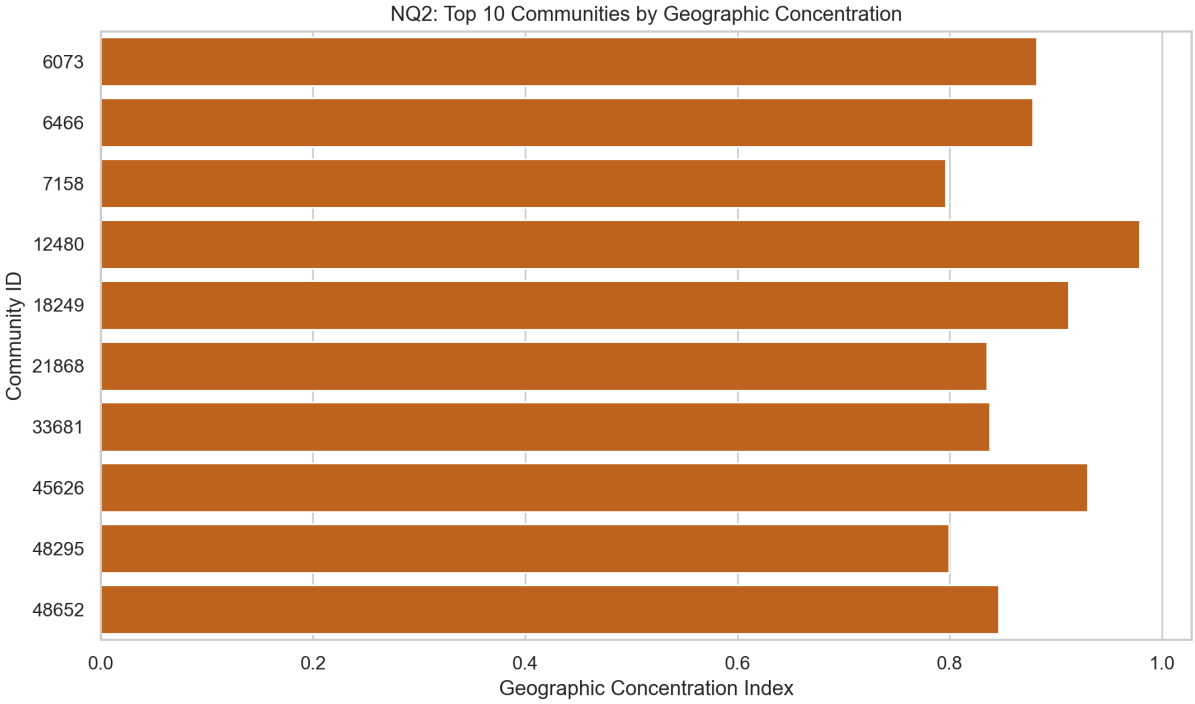

Figure 9: NQ2: Top communities by geographic concentration index.

Key findings (`n2_summary.json`):

- Communities with size ≥ 25: **19**
- Most geographically concentrated community: **12480** with GCI **0.979**
- Least concentrated community: **7449** with GCI **0.171**
- Selected projection threshold: `min_reviews = 10`

Analytical interpretation:

- GCI spans a wide range (approximately **0.17 to 0.98**), which means the social graph contains both highly local and highly cross-region communities.
- The most concentrated communities are nearly single-state clusters (for example community 12480 is AB-dominant), whereas the least concentrated communities distribute review activity across multiple states with no dominant geography.
- Across communities, top reviewed categories repeatedly include Restaurants, Food, and Nightlife, implying that **community differentiation is more geographic than categorical**. In other words, members in different communities often consume similar category types but in different places.

Business insights:

- Geo-targeted campaigns should prioritize high-GCI communities because local message relevance is likely to be strongest.
- Cross-state expansion and referral strategies are better aligned to low-GCI communities where social influence is less geographically concentrated.

NQ3: Node Similarity (Jaccard) for Market Saturation

Node Similarity is run on business nodes via shared reviewer neighborhoods. Threshold starts at 5 common reviewers and falls back to 4 or 3 if needed. Results include top-5 most saturated and top-5 most fragmented city-category combinations, excluding combinations with fewer than 5 businesses.

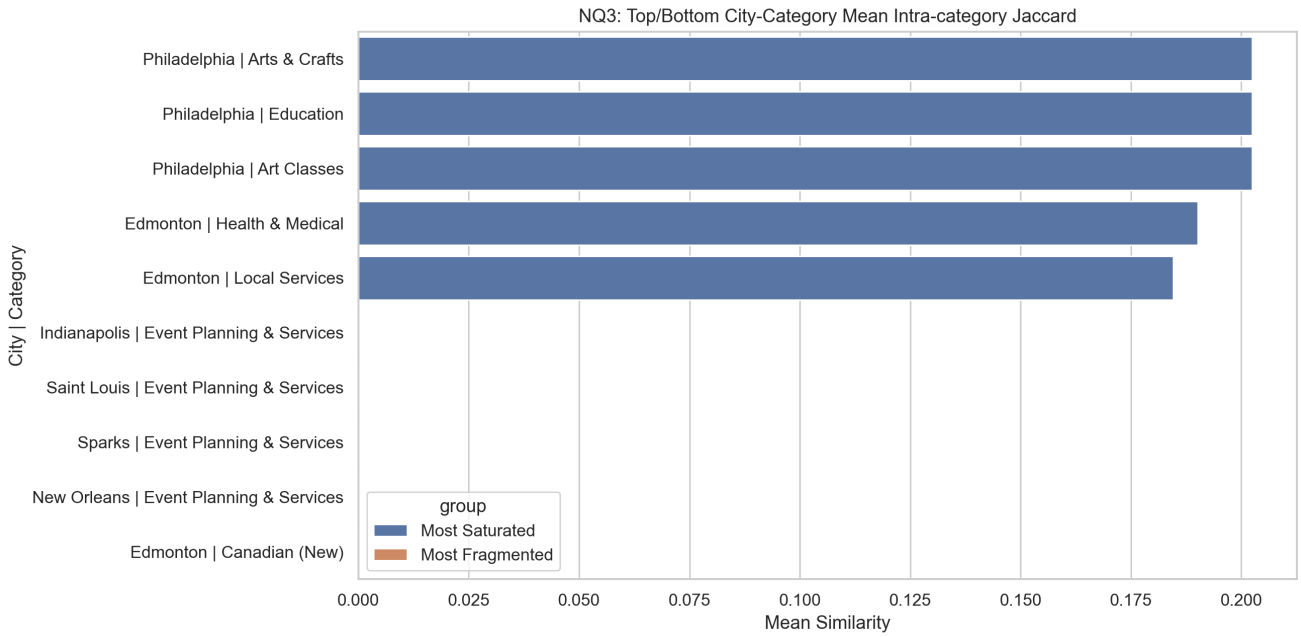

Figure 10: NQ3: Most saturated and most fragmented city-category combinations.

Key findings (`n3_summary.json`):

- NodeSimilarity configuration used: `top_k=5`, `min_common_reviewers=5`
- Most saturated combo: **Philadelphia — Arts & Crafts** (mean similarity **0.203**)
- Most fragmented combo: **Indianapolis — Event Planning & Services** (mean similarity **0.000**)

The script also computes mean stars, star spread, and mean review count for the most and least saturated combinations.

Analytical interpretation:

- Saturation (high intra-category audience overlap) appears in narrow pockets. Example: Philadelphia — Arts & Crafts has mean similarity **0.203**.
- Many city-category pairs have similarity near zero, including several Event Planning & Services combinations. This indicates fragmented demand where different businesses draw largely distinct reviewers.
- Comparing the extreme pairs in `n3_extreme_combo_business_metrics.csv`:
  - Saturated pair (Philadelphia — Arts & Crafts): mean stars **4.08**, star standard deviation **0.845**, mean review count **112.9**.
  - Fragmented pair (Indianapolis — Event Planning & Services): mean stars **4.13**, star standard deviation **0.701**, mean review count **188.0**.
- In this sample, the saturated market has **higher rating variance but lower average review volume**. This supports the interpretation that direct audience overlap can intensify competitive differentiation while limiting per-business attention.

Business insights:

- Entering saturated categories requires sharper positioning because consumers compare overlapping options directly.
- Entering fragmented categories may benefit from local niche capture and channel-specific acquisition rather than direct head-to-head brand competition.

NQ4: Betweenness vs Degree Centrality

Betweenness and degree are computed on the social graph. The top-20 overlap is reported, and high-betweenness-low-degree users are compared against high-degree users on review count, distinct cities reviewed, and distinct categories reviewed.

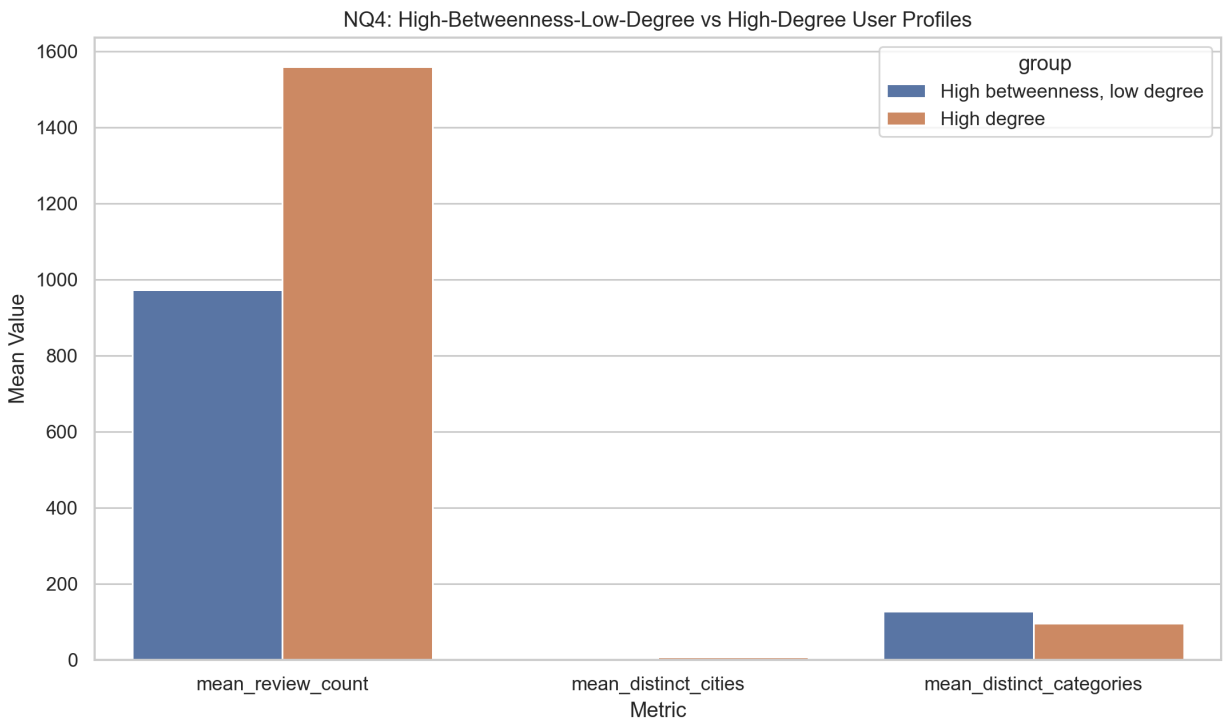

Figure 11: NQ4: Group-level comparison for centrality-defined user sets.

Key findings (`n4_summary.json`):

- Top-20 overlap between betweenness and degree lists: **16** users
- High-betweenness-but-not-top-degree users: **4**
- These users show lower mean review count (973.5 vs 1559.25) but broader category spread (128.25 vs 96.2) than high-degree users.

Analytical interpretation:

- The 16/20 overlap confirms that many hubs are also bridges, but the 4 non-overlapping high-betweenness users are analytically important.
- Those bridge users are **Isabella, Marcia, Aimee, and Ophelia** (present in top betweenness but not top degree). Their profile suggests connector behavior: fewer total links than hub users, but links that cross community boundaries.
- Compared with high-degree users, they have:
  - lower activity volume (mean review count 973.5 vs 1559.25),
  - lower geographic spread (4.75 vs 7.30 cities),

  - higher category breadth (128.25 vs 96.2 categories).

- This pattern indicates **cross-interest bridges**: users who connect topic spaces more than they maximize local neighborhood size.

Business insights:

- Degree alone is insufficient for influence seeding. Campaigns should include betweenness-heavy connectors for cross-community diffusion.
- Category-diverse connectors are valuable for introducing new categories or cross-selling between adjacent verticals.

NQ5: Link Prediction Pipeline for Recommendations

NQ5 implementation is present in `queries/p2_neo4j.py` and builds a per-user chronological train/test split over `REVIEWED_BIZ` edges before running a GDS link prediction pipeline.

At report freeze time, NQ5 remained **incomplete** due GDS-version procedure/configuration incompatibilities in the local Neo4j setup. The latest failure recorded in `report/p2/data/neo4j/n5_summary.json` is:

- `Unexpected configuration key: undirectedRelationshipTypes` during `gds.graph.project.cypher`

Therefore, stable NQ5 evaluation artifacts (AUC-ROC/Precision@10 and recommendation samples) are not included in this submission snapshot.

4 Predictive Modelling

Objective and Data Scope

Goal: predict the number of useful votes for each review (regression target: `target_useful`).

Data source is the shared materialized subset (`shared_v1`) used throughout Part 2. The feature build metadata confirms:

- dataset rows: **437,579** reviews
- dataset columns: **30**
- source DB: `yelp_db_subset_shared`

Subset Distribution Diagnostics (Exported)

The training script exports distribution diagnostics for report traceability:

- `data_bucket_distribution_matrix.csv`
- `data_distribution_summary.json`

From `data_distribution_summary.json`, the target is strongly right-skewed:

- mean useful votes: **1.471**
- median: **0**
- p95: **6**
- p99: **14**
- max: **539**

Test-set stratification (required buckets) from `data_bucket_distribution_matrix.csv`:

- 0: **45,246** (51.70%)
- 1–5: **37,045** (42.33%)
- 6–20: **4,854** (5.55%)
- 21+: **371** (0.42%)

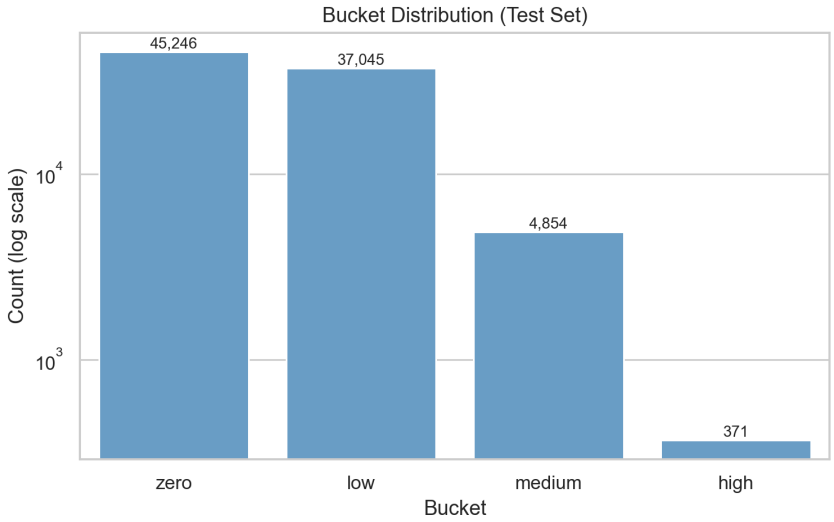

Figure 12: Predictive modelling: test-set bucket distribution (log scale).

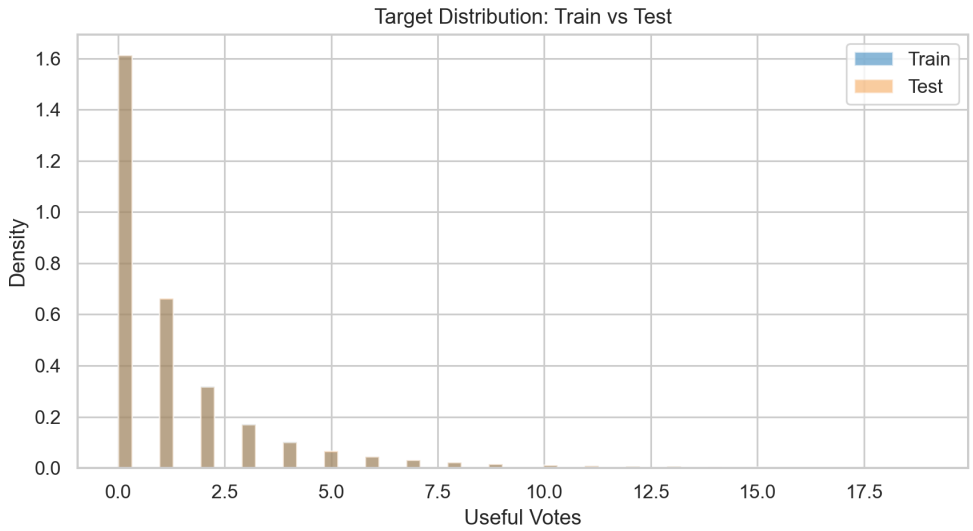

Figure 13: Predictive modelling: train vs test target distribution.

Feature Engineering (with Source/Formula)

The assignment requires review-level, user-level, and graph-derived features. The final model matrix uses 28 numeric features (18 base + engineered interactions/transforms). Table below summarizes representative features and construction.

Feature	Source	Definition / Formula
`review_stars`	Mongo reviews	Star rating of the review.
`review_text_len`	Mongo reviews	Character length of review text.
`review_word_count`	Mongo reviews	Token count in review text.
`review_recency_days`	Mongo reviews	Days from review date to latest review date in subset.
`user_review_count`	Mongo users	Total historical reviews by user profile.
`user_tenure_days`	Mongo users + review date	Review date minus user `yelping_since`.
`user_elite_years_count`	Mongo users	Number of elite years listed for user.
`user_friends_count`	Mongo users	Friend-list length after subset pruning.
`business_review_count`	Mongo businesses	Historical business review volume.
`business_average_stars`	Mongo businesses	Profile-level mean business stars.
`graph_user_degree`	Neo4j graph	User degree in FRIENDS_WITH social graph.
`graph_user_community_size`	Neo4j NQ2 output	Louvain community size linked via user community id.
`graph_user_community_gci`	Neo4j NQ2 output	Geographic concentration index of user community.
`graph_business_city_category_similarity`	Neo4j NQ3 output	Mean city-category intra-category reviewer similarity (saturation proxy).
`user_reviews_per_tenure_day`	Engineered	`user_review_count / (user_tenure_days + 1)`.
`review_star_x_business_avg`	Engineered	Interaction: review stars × business average stars.
`business_popularity_x_saturation`	Engineered	`business_review_count` × `graph_business_city_category_similarity`.

Table 1: Predictive feature examples with data sources and formulas.

Graph-feature rationale (assignment requirement):

- `graph_user_degree`: users with denser social neighborhoods plausibly produce reviews with broader exposure.
- `graph_user_community_gci`: community structure/geographic concentration can influence downstream visibility and relevance.
- `graph_business_city_category_similarity`: market saturation affects how comparable and discoverable reviews are within category-city niches.

Modelling Strategy and Methods

Train/test split follows the required stratification buckets: {0, 1–5, 6–20, 21+}. Candidate models were evaluated in an auto-selection pipeline:

- Histogram Gradient Boosting regressors (standard and log-target variants)
- Poisson-loss gradient boosting variants
- Tweedie/Poisson GLM baselines
- Fast random forest baseline
- Hurdle model (two-stage: classify useful>0, then regress positive counts)
- Blend candidates combining complementary predictors

Model objective selection used minimum RMSE on the held-out test set. Feature importance is reported via permutation importance (on sampled test rows) for interpretability.

Model Comparison and Final Selection

Model matrix was exported to `model_metrics_matrix_2d.csv` and visualized using `model_comparison_metrics_lines`

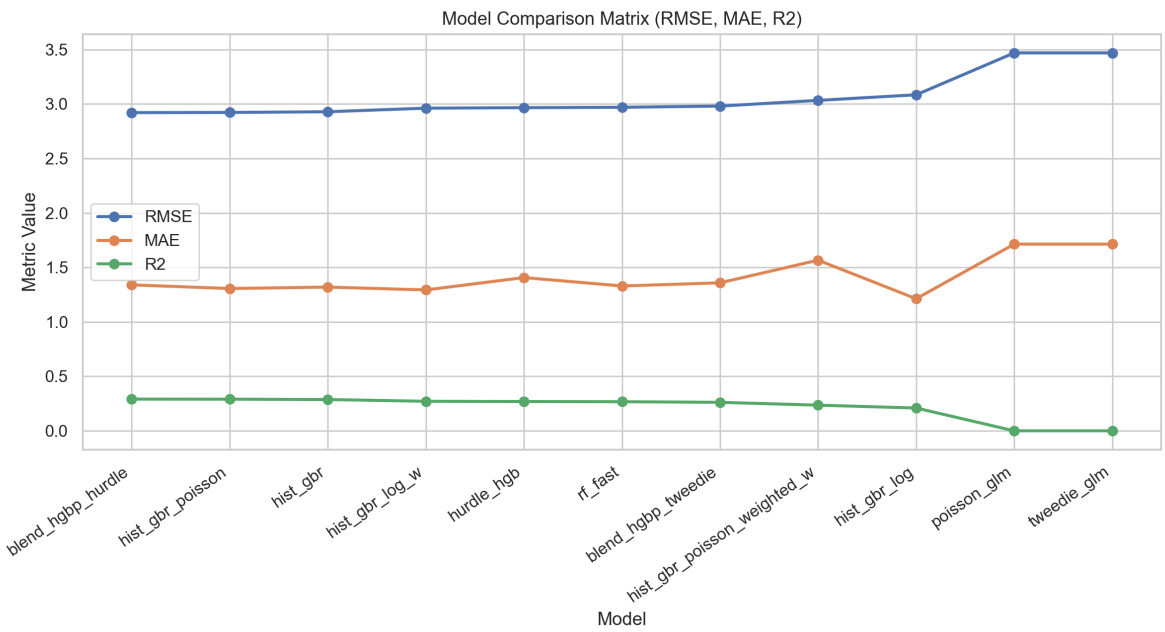

Figure 14: Predictive modelling: RMSE/MAE/R^2 across candidate models.

Best model selected: **blend_hgbp_hurdle**

- RMSE: **2.9204**
- MAE: **1.3394**
- R^2: **0.2911**

The blend achieved the lowest RMSE among all candidates, slightly improving over the best single model (`hist_gbr_poisson`).

Required Bucket-wise Evaluation

From `bucket_metrics.csv` (required groups: 0, 1–5, 6+):

- **0 useful votes**: RMSE 1.3312, MAE 0.8937
- **1–5 useful votes**: RMSE 1.7067, MAE 1.1706
- **6+ useful votes**: RMSE 10.3371, MAE 6.3953

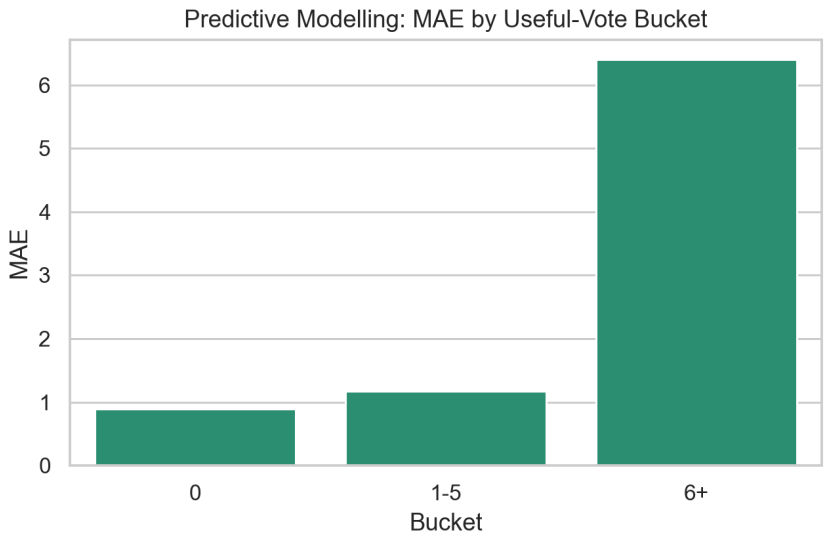

Figure 15: Predictive modelling: MAE by useful-vote bucket.

Interpretation:

- Performance is strongest in the dominant low-count regime (0 and 1–5 buckets).
- Error increases sharply in 6+ due to extreme sparsity and heavy-tail variance, which is expected in useful-vote prediction.

Prediction Diagnostics

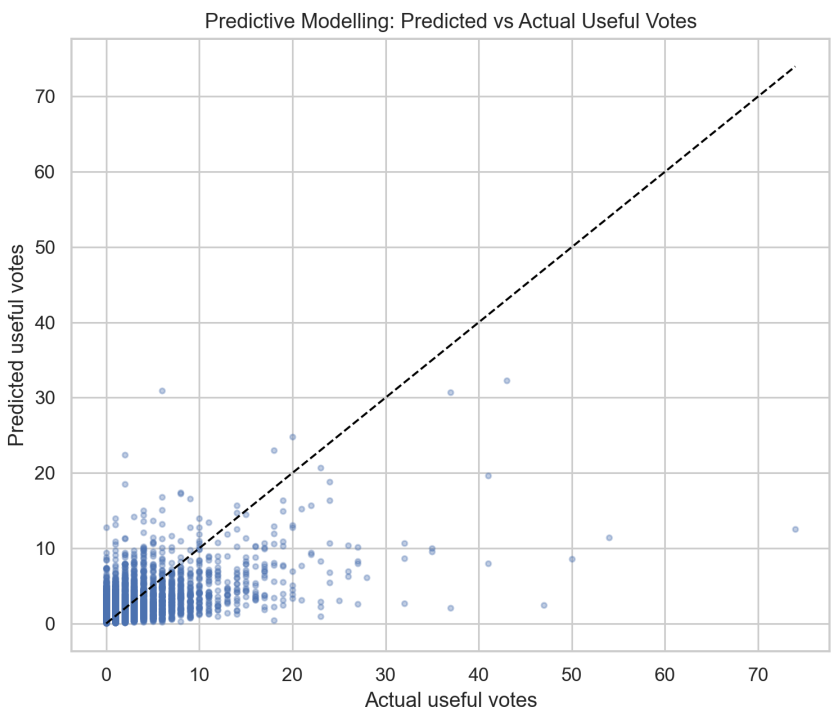

Figure 16: Predictive modelling: predicted vs actual useful votes.

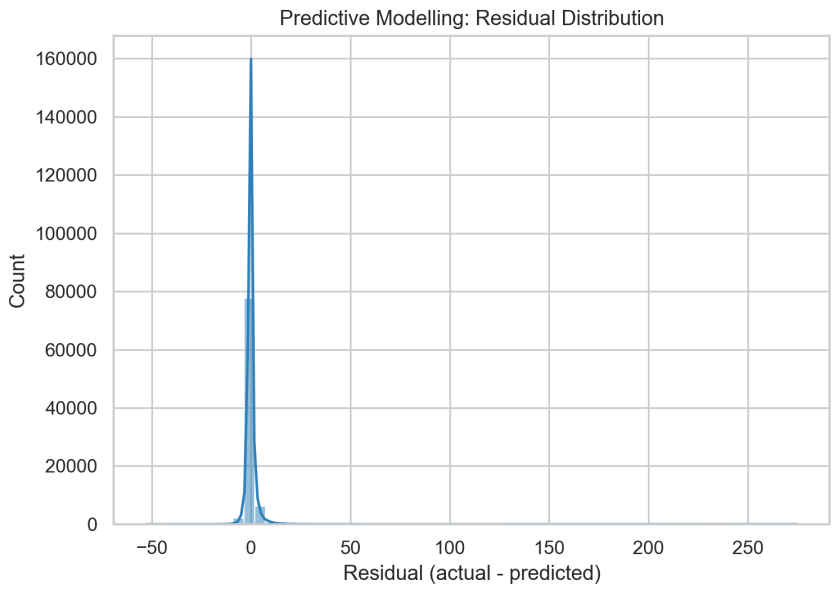

Figure 17: Predictive modelling: residual distribution.

The scatter and residual plots show central-mass fit quality but persistent tail underfit in high useful-vote reviews, consistent with bucket-wise results.

Feature Importance and Discussion

Top-3 predictive features (permutation importance, `feature_importance.csv`):

- `review_recency_days` (0.2156)
- `business_review_count` (0.1198)
- `review_text_len` (0.1071)

Data-supported interpretation:

- **Recency** dominates, suggesting platform attention dynamics are time-sensitive (newer/temporally positioned reviews attract different engagement levels).
- **Business review volume** indicates exposure effects: reviews for more visible businesses tend to receive more interactions.
- **Text length** captures informational richness; longer reviews often provide more evaluative content and can attract more useful votes.

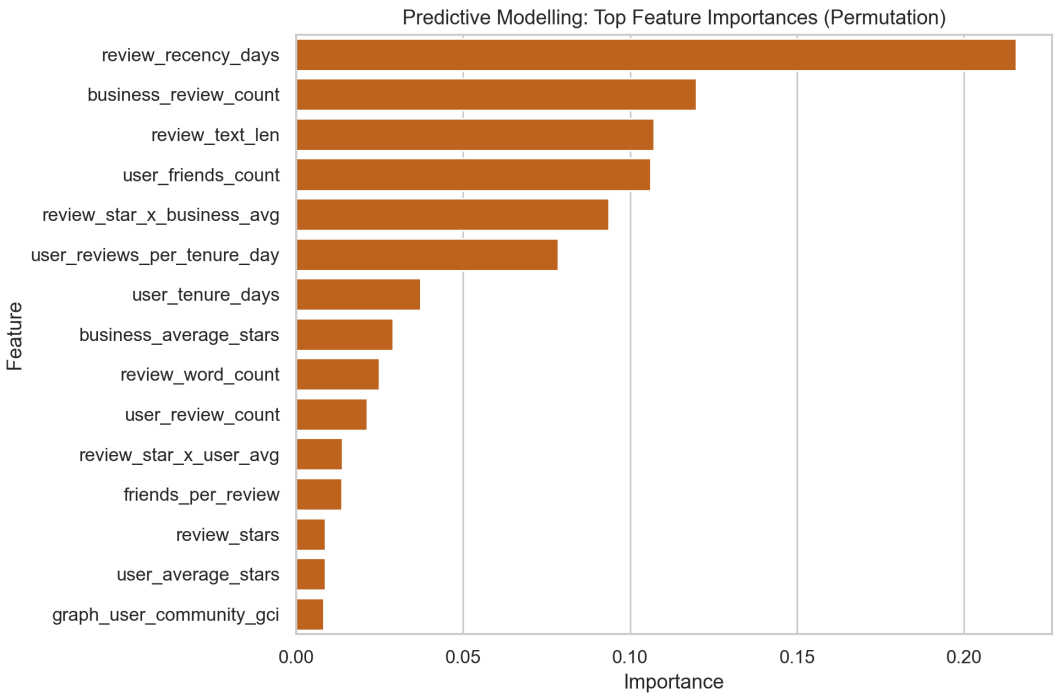

Figure 18: Predictive modelling: top feature importances.

Limitation and Potential Consequence

Limitation: the target is highly imbalanced with very few extreme high-useful instances (21+), and even fewer very large values. This limits tail calibration and reduces reliability for ranking the very top reviews.

Potential consequence if deployed for ranking: an under-calibrated model may systematically under-rank niche but high-value reviews (especially from smaller businesses or less represented communities), reinforcing popularity bias toward already visible entities.

5 Query Text Files

The full Part 2 query text files (MongoDB and Neo4j) are included separately in the submitted ZIP package.

- Mongo queries: `queries/mongo_part2_queries.txt`
- Neo4j queries: `queries/cypher_part2_queries.txt`